

Linear Algebra - CS - 3A & 3B

ProblemSheet: Eigenvalues, Eigenvectors and Diagonalization

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There are two types of questions; one "Mechanical" which requires you to do computations and other "Conceptual", which requires you to respond question logically using theorem and logical implications. Attempt accordingly.

A Eigen Values and Eigen Vectors

1. **(Mechanical)** For $E = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$, find eigenvalues and eigenvectors.
Verify that eigenvectors corresponding to distinct eigenvalues are orthogonal. Note orthogonal vectors are perpendicular, which make 90 degree angle and their dot product is zero i.e. $\mathbf{u} \cdot \mathbf{v} = 0$.
2. **(Conceptual)** Let A be a 2×2 matrix with eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. What are the eigenvalues of A^2 ? Of A^{-1} (if invertible)? Of $A + 3I$?
3. **(Conceptual)** Show that a 2×2 real matrix with a repeated eigenvalue λ is diagonalizable if and only if it is already λI .
4. **(Conceptual)** Suppose v is an eigenvector of both A and B . Is v necessarily an eigenvector of AB ? Prove or give a counterexample.

5. **(Mechanical)** Compute eigenvalues and eigenvectors for

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

6. **(Conceptual)** If A is a 3×3 matrix with eigenvalues $1, 2, 3$, what are the eigenvalues of A^T ? What about $A^2 - 2A + I$?
7. **(Mechanical)** Find the eigenvalues and a basis for each eigenspace of

$$D = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Is D defective? Note that matrix A is called defective if their all eigenvectors are not linearly independent, means non-diagonalizable.

8. **(Conceptual)** Prove that if λ is an eigenvalue of an orthogonal matrix Q , then $|\lambda| = 1$. Note that matrix A is called orthogonal if $A^T = A^{-1}$

B Diagonalization

1. **(Mechanical)** Diagonalize the matrix $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. That is, find P and D such that $A = PDP^{-1}$.
2. **(Mechanical)** Find P and D for $C = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$ and compute C^{10} using the diagonalization.
3. **(Conceptual)** Can a non-invertible 2×2 matrix be diagonalizable? If yes, give an example. If no, explain why.

4. **(Mechanical)** Determine whether $B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable.

If so, diagonalize it.

5. **(Conceptual)** Suppose A and B are diagonalizable 2×2 matrices with the same eigenvectors. Does it follow that $AB = BA$? Explain.

6. **(Mechanical)** Diagonalize the symmetric matrix

$$D = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

with an orthogonal matrix P (i.e., find P orthogonal and D diagonal).